



Demand Forecasting — Executive Summary

Gradient Boosting · Recursive 8-Week Forecast · Walk-Forward Validation · Jan–Feb 2022

3.0%

SMAPE accuracy



3.0%

SMAPE

Beats Seasonal Naive (4.2%)



14.6

MAE (units/week)

Per supermarket-SKU combination



+6–10%

Promo Uplift

Demand spike during deal weeks



–16%

Jan–Feb Demand

Soft patch vs Dec across all SKUs

What Drives Demand?



Recent History (Lags)

Lag features (1–52 wks) are the single biggest predictor. Demand is sticky — this week's sales strongly predict next week's.



Seasonality

Clear Q4 peaks and mid-year dips across all SKUs. Cyclical sin/cos encoding captures the annual rhythm without discontinuity.



Promotions (+6–10% uplift)

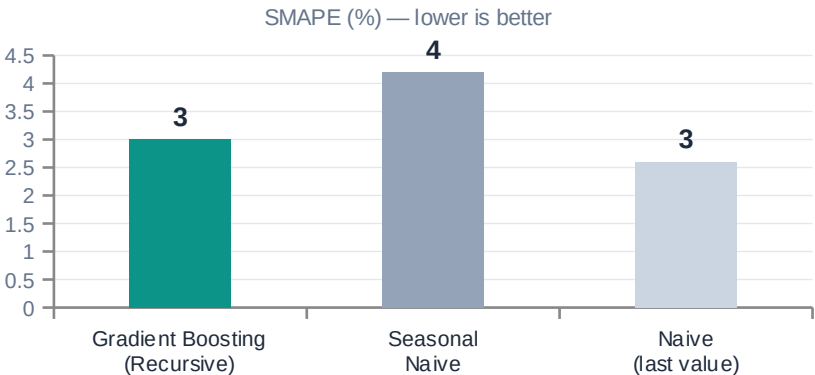
15 promo events across 6 combos. FreshMart Eggs and GreenBasket Bread show the strongest uplifts when deals run.



Slow Upward Trend

~2–3% annual baseline growth across SKUs — consistent with steady store footfall growth over 2019–2021.

How Good Is the Forecast?



MAE (units/week)

14.6

Gradient Boosting

20.4

Seasonal Naive

12.5

Naive

Why SMAPE?

Symmetric — over-forecast (write-off) and under-forecast (stock-out) are penalised equally, matching the business cost structure. Scale-independent — comparable across Milk (350 u/wk) and Bread (270 u/wk) without distortion. Interpretable — 3.0% means ~9 units wrong per 300 ordered.

Forecast & Business Implications

SKU	03-Jan	10-Jan	24-Jan	07-Feb	21-Feb
Free Range Eggs	300	279	261	263	258
Organic Milk	336	331	292	289	274
Wheat Bread	295	284	263	259	253

Avg units/week across all 3 supermarkets



Trim Safety Stock

At 3.0% SMAPE, ~9 units buffer covers a 300-unit order. Planners can safely reduce over-stocking and cut write-offs.



Plan for Jan–Feb Softness

Avg demand falls from 311 → 261 units/week across Jan. Scale deliveries down mid-January to avoid surplus.



Watch Organic Milk Closely

Highest volume (274–407 u/wk) and widest forecast swing. Stock-out risk is greatest here — monitor weekly.